The COMS Randomized Trial of Iodine 125 Brachytherapy for Choroidal Melanoma, II: Characteristics of Patients Enrolled and Not Enrolled

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The Collaborative Ocular Melanoma Study Group

Objectives: To describe characteristics of patients evaluated for the Collaborative Ocular Melanoma Study (COMS) randomized clinical trial of iodine 125 brachytherapy for choroidal melanoma by enrollment status, and to compare characteristics of patients enrolled with those of patients with tumors of eligible size who did not enroll in order to assess the extent to which findings from the clinical trial can be generalized to future patients.

Methods: For all patients diagnosed with choroidal melanoma and evaluated for the clinical trial at COMS centers from November 1986 through July 31, 1998, selected data were transmitted to the COMS Coordinating Center, Baltimore, Md, where they were integrated and analyzed. Data included ophthalmic and medical history, examination findings, and visual acuity measurements recorded prior to enrollment; standardized A- and B-scan echographic examination findings; and wide-angle fundus photographs and fluorescein angiograms.

Results: Of 8712 patients with choroidal melanoma, 5046 had tumors of eligible size; of these, 2882 (57%) were eligible for enrollment, and 1317 (46% of eligible patients, 26% of patients with tumors of eligible size) enrolled. Most differences between eligible and ineligible patients corresponded to eligibility and exclusion criteria. However, ineligible patients were older and had thicker tumors than eligible patients. Eligible patients who enrolled were slightly older and had larger tumors than those who did not enroll. Nearly half (48%) of enrolled patients had choroidal melanoma with the apex located temporal to the fovea, compared with 40% of eligible patients not enrolled and 29% of ineligible patients.

Conclusions: This trial was designed to yield internally valid treatment comparisons through random assignment to treatment at time of enrollment. Information from this and other studies document that enrolled patients were similar to other patients with choroidal melanoma who were treated with ^{125}I brachytherapy. These findings support the external validity of the trial and applicability of treatment findings to all patients who meet the criteria used to judge eligibility for the trial.

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ALTHOUGH RARE, choroidal melanoma is the most common primary intraocular cancer in adults.[1] Enucleation has been the accepted treatment for primary choroidal melanoma for more than a century and remains so for large tumors. During the decades that preceded initiation of the present study, other methods of treatment were proposed as alternatives to enucleation of eyes containing smaller choroidal melanoma, with the goal of retaining the eye and possibly useful vision.[2-15] Apart from the desire of many patients to retain their eyes, other factors that stimulated a search for alternatives to enucleation included diagnostic uncertainty, particularly in the case of smaller tumors, as fine-needle aspiration of choroidal melanoma typically is not performed, and findings reported by Zimmerman et al,[16,17] which the authors interpreted to implicate enucleation as a cause of distant metastasis. These issues are discussed more fully in a separate report.[18]

See also pages 969 and 1067

At the time the Collaborative Ocular Melanoma Study (COMS) was designed, radiotherapy was the most promising eye-conserving treatment, whether delivered by means of charged particles[19-22] or an episcleral plaque to which radioactive material was affixed.[23-25] Thus, the COMS was designed to evaluate radiotherapy vs enucleation in the subset of patients with choroidal melanoma for whom radiotherapy would be considered by ophthalmologists and patients if it would not reduce survival or increase com-

PATIENTS AND METHODS

Descriptions of many aspects of the COMS design and methods have been published.[26,28-36] The *COMS Manual of Procedures*[37] and the *COMS Forms Book*[38] are available.

INITIAL EVALUATION AND CASE REPORTING

The Data and Safety Monitoring Committee appointed by the director of the National Eye Institute, Bethesda, Md, approved the study design and methods on August 27, 1986, prior to initiation of patient evaluation and enrollment. In addition, the institutional review board of each participating institution reviewed and approved the COMS protocol and the consent forms used at the respective clinical center prior to initiation of data reporting and patient enrollment at individual centers.

Patients with suspected choroidal melanoma were evaluated at 1 of 43 COMS clinical centers for confirmation of the diagnosis and evaluation of eligibility for one of the COMS randomized trials. All patients for whom the diagnosis of choroidal melanoma was confirmed clinically by a COMS ophthalmologist during the enrollment period were reported to the COMS Coordinating Center (Baltimore, Md) but were identified only by study numbers and code names assigned at the local clinical centers.

A brief medical and ocular history was elicited before the patient underwent a detailed ophthalmic examination by a COMS ophthalmologist. The history and examination addressed the eligibility of the patient and provided baseline descriptive data. Best-corrected visual acuity was measured on a Bailey-Lovie chart according to a standard protocol.[37] As part of the ophthalmic examination, the ophthalmologist estimated the tumor size and provided a detailed description of the tumor location based on distances from landmark intraocular structures. In particular, the ophthalmologist noted the location relative to the optic disc when assessing eligibility for enrollment in the COMS clinical trial of ^{125}I brachytherapy. Estimates of apical height and longest basal diameter of the choroidal melanoma took account of both the clinical observations and local measurements from A-scan echograms. Some patients who were discovered to be ineligible early in the evaluation process or who refused to consider enrollment in the COMS had examinations or procedures that deviated from the COMS protocol. Information from the screening examination was sometimes incomplete for these patients.

Each patient judged to be eligible for the COMS on the basis of the ophthalmologic examination and medical history was referred to the COMS radiation oncologist for discussion of radiation issues. The radiation oncologist, a medical oncologist, or an internist elicited a detailed medical history, performed a cancer-oriented physical examination, and ordered and interpreted laboratory studies and an anterior-posterior x-ray film of the chest. A computed tomographic scan, magnetic resonance images, or biopsy of the liver or other organs was required by the COMS protocol whenever necessary to rule out metastatic melanoma or other neoplasms following defined abnormalities in tests of liver function.

For all patients who enrolled in the clinical trial and some early eligible patients who did not enroll, the Echography Center (currently located in Mars Hill, NC) reviewed standardized A-scan and contact B-scan echograms for consistency with the diagnosis and evidence of extrascleral extension, measured the apical height of the tumor, and classified tumor configuration and internal reflectivity according to specified criteria.[36] Wide-angle photographs and a fluorescein angiogram with frames taken of all tumor margins whenever possible were reviewed at the COMS Photograph Reading Center (Iowa City, Iowa) for consistency with the diagnosis of choroidal melanoma. The clinical center ophthalmologist was notified whenever extrascleral extension was noted at the Echography Center and whenever the diagnosis was questioned by personnel at either of these 2 resource centers.

ELIGIBILITY FOR RANDOMIZED TRIAL OF ^{125}I BRACHYTHERAPY

From November 1986 until July 31, 1998, patients with choroidal melanoma were evaluated for eligibility for the COMS trial of ^{125}I brachytherapy. To be considered for this randomized trial, the patient had to have choroidal melanoma from 2.5 to 10.0 mm in apical height and no more than 16.0 mm in longest basal diameter. (Until November 1990, the lower and upper limits on apical height were 3.1 and 8.0 mm, respectively.) Peripapillary tumors (those with the proximal tumor border 2.0 mm or closer to the optic disc) were eligible only when the tumor was contained within a 90° angle, with the apex at the center of the optic

plications. The COMS Group elected to conduct 2 randomized trials of radiotherapy for choroidal melanoma. In each trial, the control arm consisted of patients undergoing standard enucleation alone. The primary outcome of each trial was death from any cause; death attributable to melanoma metastasis was the principal secondary outcome. In the randomized trial for large choroidal melanoma, the radiotherapy arm consisted of patients undergoing preenucleation external beam irradiation. Initial findings from that trial have been published.[26-28] In the COMS clinical trial that is the focus of this report, iodine 125 brachytherapy was used. The rationale for selecting ^{125}I as the isotope and a radioactive plaque delivery system has been presented elsewhere.[29,30] A major goal was to evaluate a method of delivering radiotherapy that could be standardized and made available at many institutions, both within the COMS organization during the course of the clinical trial and throughout the medical community if it provided survival rates comparable or superior to those for enucleation. In addition, this method of radiotherapy delivery was expected to be applicable to most patients with choroidal melanoma for whom an alternative to enucleation would be considered by their ophthalmologists.

Accrual of patients to the COMS clinical trial of ^{125}I brachytherapy was completed on July 31, 1998. The purpose of this report is to describe the characteristics of the study population and to compare them with characteristics of patients with tumors of eligible size who were not enrolled. This comparison permits assessment of the external validity of the trial, that is, the extent to which patients who participated represent the target populadisc, and when the enrolling ophthalmologist was confident that a radioactive episcleral plaque could be placed to cover the entire base of the tumor and a 2-mm margin beyond the tumor borders apart from the border proximal to the optic disc. Patients whose tumors were contiguous with the optic disc were ineligible.

Eligible patients had a primary choroidal melanoma in 1 eye only; were aged 21 years or older; had no other primary tumor and no history of cancer other than noninvasive nonmelanotic skin cancer or carcinoma in situ of the uterine cervix; had no coexisting disease that threatened survival for 5 years or longer; were judged by the examining oncologist or internist to be free of metastatic melanoma; had best-corrected visual acuity in the fellow eye of 20/200 or better; and were able to give informed consent, willing to adhere to local radiation safety guidelines, and had no condition that would prevent them from returning for posttreatment follow-up examinations. Previous treatment for choroidal or ciliary body melanoma in either eye, treatment for any condition secondary to the tumor, or fine-needle aspiration biopsy of the melanoma rendered a patient ineligible for the clinical trial. Patients who had extrascleral tumor extension of 2.0 mm or greater thickness detected during echography or clinical examination, diffuse, ring, or multifocal tumors, or tumors that were judged to be located primarily in the ciliary body also were ineligible. Additional criteria for eligibility are described elsewhere.[18,37]

INITIAL TREATMENT

Patients who were judged eligible after completion of all baseline evaluations were invited to participate in the trial. Those who gave signed consent were enrolled by means of a telephone call from the local clinic coordinator and ophthalmologist to the Coordinating Center. During the call, eligibility was reviewed, and the random assignment to ^{125}I brachytherapy or enucleation was communicated to the clinical center personnel by Coordinating Center personnel.

Eligible patients who did not enroll selected the management approach in consultation with their ophthalmologists. The treatment selected was reported to the Coordinating Center for most eligible patients not enrolled. Reporting of treatment selected by patients not enrolled because they were judged ineligible was initiated in September 1989.

DATA COLLECTION, MANAGEMENT, AND ANALYSIS

Data from clinical evaluations and from review of echograms were recorded on standard forms[38] and forwarded to the Coordinating Center for transcription to computer files, automated and manual edits for consistency and protocol adherence, and data analysis. At several points in the course of the COMS, the investigators evaluated individual data collection items with respect to their contributions to the primary and secondary outcomes of the trials and to improved knowledge of choroidal melanoma. Some changes were made at the time of each of these reviews. Therefore, some information was collected during only part of the enrollment period and is available for only a subset of enrolled patients. Less information was requested for patients who did not enroll than was required for those who enrolled. Data missing for these 2 reasons are categorized as "not available" in the tables. Eligibility of patients who enrolled in the trial and reasons for ineligibility and for nonparticipation of eligible patients were confirmed during review of clinic records and chart notes as part of routine visits by COMS clinic monitors to participating clinical centers.

Only 1 record was retained in the database for each patient reported to the COMS. Comparisons were performed twice yearly to assure that each patient reported by more than 1 center was counted only once.[26] The most recent eligibility and enrollment status of each patient reported was retained for all data analyses.

Data reported herein are from baseline examinations, central reviews of baseline materials, and any corrections to baseline data that were received at the Coordinating Center by September 30, 2000. Data displays are primarily descriptive in nature. Statistical tests were used to determine whether there were informative differences between patients who were and were not eligible for the trial with respect to characteristics of patients and choroidal melanoma reported for both groups and to compare eligible patients who enrolled in the trial with those who did not. The Wilcoxon rank sum test was used to compare distributions of continuous variables.[39] The χ^2 test for trend in ordered categories[40] was used to compare distributions of categorical variables in 3 or more qualitatively or quantitatively ordered categories. The χ^2 test for homogeneity[39] was used to compare dichotomous distributions and distributions in which the categories have no inherent qualitative or quantitative order.

tion who have choroidal melanoma and for whom treatment with ^{125}I brachytherapy could be considered, and, thus, the ability to generalize the findings to future patients. Survival outcomes as of September 30, 2000, are reported in a separate article.[18]

RESULTS

REASONS FOR INELIGIBILITY AND NOT ENROLLING

As of July 31, 1998, 8712 patients with choroidal melanoma of all sizes had been evaluated at COMS clinical centers and reported to the Coordinating Center; 5046 patients (58%) had tumors that met the size requirements for this clinical trial. Of 2164 patients with tumors of appropriate size who were ineligible for the trial (43%), 470 (22%) failed to meet 2 or more of the eligibility criteria. The most common reasons for exclusion of the remaining 1694 patients were proximity (183 patients) or contiguity (495 patients) of the tumor to the optic disc, which precluded protocol treatment with an episcleral radioactive plaque (40%); 1 or more other primary cancers (332 patients, 20%); and melanoma that was predominantly in the ciliary body (193 patients, 11%). These reasons were also those most often reported for patients ineligible for 2 or more reasons; the corresponding numbers and percentages for these 470 patients were 179 (38%), 166 (35%), and 136 (29%). The contralateral eye of 5 patients otherwise eligible for this clinical trial had an earlier or simultaneous diagnosis of choroidal melanoma; 5 more patients with bilateral choroidal

Table 1. Demographic Characteristics of Eligible and Ineligible Patients

Characteristic	No. (%) of Patients			
	Eligible Patients			
	Enrolled (n = 1317)	Not Enrolled (n = 1565)	Total Eligible (n = 2882)	Ineligible Patients (n = 2164)
Age, y				
<40	125 (9)	176 (11)	301 (10)	162 (7)
40-49	178 (14)	262 (17)	440 (15)	236 (11)
50-59	268 (20)	328 (21)	596 (21)	346 (16)
60-69	408 (31)	382 (24)	790 (27)	526 (24)
70-79	282 (21)	320 (20)	602 (21)	622 (29)
≥80	56 (4)	91 (6)	147 (5)	271 (13)
Not reported	0	6	6	1
Mean age, y	60	59	59	64
Sex				
Men	665 (50)	813 (52)	1478 (51)	1056 (49)
Women	652 (50)	746 (48)	1398 (49)	1107 (51)
Not reported	0	6	6	1
Race/ethnicity				
White, not Hispanic	1289 (98)	1534 (98)	2823 (98)	2117 (98)
Hispanic	14 (1)	14 (1)	28 (1)	29 (1)
Black, not Hispanic	8 (1)	7 (<1)	15 (1)	13 (1)
Asian or Pacific Islander	5 (<1)	4 (<1)	9 (<1)	3 (<1)
Native American	1 (<1)	0	1 (<1)	1 (<1)
Not reported	0	6	6	1

melanoma were ineligible for additional reasons. Twenty-one patients were diagnosed as having metastatic melanoma at time of evaluation for the COMS. Twenty-four patients were ineligible because they were younger than 21 years; young age was the only reason for ineligibility for 15 of these 24 patients.

Of 2882 patients judged eligible for enrollment, 1317 (46% of eligible patients; 26% of all patients with choroidal melanoma of appropriate size) gave signed consent, enrolled, and were assigned randomly to enucleation or to an ^{125}I radioactive plaque. Management preferences of the patient or of the referring or COMS ophthalmologist accounted for the decision of 1143 (73%) of the 1565 eligible patients not enrolled. Unwillingness of the patient to accept random assignment of treatment accounted for 345 more (22%).

INITIAL TREATMENT ELECTED BY PATIENTS NOT ENROLLED

Of the 1537 eligible patients not enrolled for whom initial treatment of the tumor was reported, 469 (31%) selected enucleation, either alone or following preenucleation radiation; 887 (58%) selected radioactive plaque, either alone or in combination with laser photocoagulation to the tumor borders; 27 (2%) had other types of eye-conserving radiotherapy; 27 (2%) had other types of treatment, and 127 (8%) initially were observed without treatment. Of the latter group, 71 (56%) had choroidal melanoma with apical heights in the range 2.5 mm to 3.0 mm.

The treatment selected was reported for 1741 ineligible patients. Among these patients, 674 (39%) had the eye enucleated, 720 (41%) had some type of eye-conserving radiotherapy, and 210 (12%) were observed without treatment initially. Of the latter group, 59 had another primary cancer, 23 had undergone treatment of the tumor prior to evaluation for the COMS, 19 had another life-threatening condition, and 6 had melanoma metastases.

BASELINE DEMOGRAPHIC CHARACTERISTICS

Demographic characteristics of the patients enrolled in the randomized trial are displayed in **Table 1**. Characteristics of eligible patients not enrolled and of patients who had choroidal melanoma that met the size criteria but who were not eligible for enrollment are given for comparison. A detailed analysis of sociodemographic and clinical factors potentially predictive of enrollment in the COMS trials is reported elsewhere.[41]

Ineligible patients were older than eligible patients (Table 1; $P<.001$), with mean ages of 64 years and 59 years, respectively. Eligible patients who enrolled were somewhat older than eligible patients who did not enroll ($P=.04$). Nearly identical numbers of men and women were in each eligibility and enrollment category (Table 1). Almost all patients with choroidal melanoma were non-Hispanic white, regardless of whether eligible for the COMS clinical trial; this finding is consistent with the epidemiology of this cancer.[42] Two American Indians evaluated for the COMS had tumors of eligible size; one was eligible and enrolled but the other was ineligible. These 2 cases were reported earlier by the COMS Group.[43]

TUMOR CHARACTERISTICS

Of the patients who enrolled in the clinical trial, 944 (72%) were evaluated at a COMS center within 30 days of the

Table 2. Characteristics of Choroidal Melanoma in Eligible and Ineligible Patients*

Characteristic	No. (%) of Patients			
	Eligible Patients			
	Enrolled (n = 1317)	Not Enrolled (n = 1565)	Total Eligible (n = 2882)	Ineligible Patients (n = 2164)
Time since initial diagnosis, d				
≤30	944 (72)	1190 (76)	2134 (74)	1572 (73)
31-180	188 (14)	170 (11)	358 (12)	254 (12)
181-365	67 (5)	56 (4)	123 (4)	78 (4)
>365	117 (9)	148 (9)	265 (9)	246 (11)
Not reported	1	1	2	13
Median time, d	9	8	8	7
Laterality				
Right eye	663 (50)	821 (52)	1484 (51)	1075 (50)
Left eye	654 (50)	744 (48)	1398 (49)	1084 (50)
Both eyes	. . .	. . .	. . .	5 (<1)
Apical height, mm				
2.5-3.0	162 (12)	297 (19)	459 (16)	294 (14)
3.1-4.0	430 (33)	472 (30)	902 (31)	613 (28)
4.1-5.0	243 (18)	268 (17)	511 (18)	372 (17)
5.1-6.0	154 (12)	199 (13)	353 (12)	288 (13)
6.1-7.0	139 (11)	138 (9)	277 (10)	233 (11)
7.1-8.0	114 (9)	104 (7)	218 (8)	192 (9)
8.1-10.0	75 (6)	87 (6)	162 (6)	171 (8)
Not reported	. . .	. . .	. . .	1
Mean height, mm	4.8	4.7	4.7	5.0
Longest basal diameter, mm				
≤8.0	185 (14)	252 (16)	437 (15)	378 (18)
8.1-10.0	275 (21)	378 (24)	653 (23)	497 (23)
10.1-12.0	371 (28)	426 (27)	797 (28)	539 (25)
12.1-14.0	276 (21)	302 (19)	578 (20)	378 (18)
14.1-16.0	210 (16)	207 (13)	417 (15)	348 (16)
Indeterminate	. . .	. . .	. . .	24
Mean diameter, mm	11.4	11.1	11.2	11.2

*Ellipses indicate not applicable.

initial diagnosis of choroidal melanoma (**Table 2**). For enrolled patients, the median interval from diagnosis to evaluation for the COMS was 9 days and from diagnosis to enrollment was 20 days. Median time from diagnosis to evaluation at a COMS clinical center was 8 days for patients who were eligible but did not enroll and 7 days for those who were ineligible.

Choroidal melanoma was diagnosed with similar frequency in right and left eyes in all groups of patients (Table 2). The mean apical height of the tumor was similar for eligible patients who enrolled (4.8 mm) and who did not enroll (4.7 mm); however, the distributions of apical height differed ($P = .001$) with patients who enrolled having tumors of greater apical height. The distributions of apical height also differed between eligible and ineligible patients ($P < .001$), with ineligible patients having thicker tumors (Table 2). Although the mean longest basal tumor diameter was similar in eligible patients who enrolled (11.4 mm) and those who did not (11.1 mm), these distributions also differed ($P = .005$), with more enrolled patients having tumors with longer basal diameters. Both the means and the distributions of longest basal tumor diameter were similar for all eligible patients combined and ineligible patients.

Findings from the ophthalmologist's description of tumor location relative to intraocular landmarks are summarized in **Table 3**. The locations of the most anterior and posterior borders of the choroidal melanoma were reported for all but 32 eligible patients. For 97% of eligible patients, regardless of the enrollment decision, the posterior border was posterior to the equator; for the remaining eligible patients, the most posterior portion of the tumor border was between the equator and the ora serrata. In contrast, of the 1291 ineligible patients for whom the information was reported, the posterior tumor border was anterior to the equator in 183 (14%). Differences also were noted between eligible and ineligible patients and between eligible patients who enrolled and those who did not enroll with respect to the location of the most anterior border of the tumor ($P < .001$ and $P = .01$, respectively). Among eligible patients who enrolled, 45% had the most anterior border of the choroidal melanoma anterior to the equator compared with 50% of eligible patients who did not enroll. The anterior tumor border was anterior to the equator in 44% of the 1284 ineligible patients for whom the information was reported; in 4% the tumor extended into the anterior chamber angle, and in 5% the tumor invaded the iris.

More ineligible patients than eligible patients had the tumor apex over the fovea (Table 3). Nearly half (48%) of the patients who enrolled had choroidal melanoma with the apex temporal to the fovea compared with eligible

Table 3. Ophthalmologic Assessment of Location of Choroidal Melanoma in Eligible and Ineligible Patients*

Characteristic	No. (%) of Patients: Eligible Patients: Enrolled (n = 1317)	Not Enrolled (n = 1565)	Total Eligible (n = 2882)	Ineligible Patients (n = 2164)
Location of posterior tumor border				
Posterior to equator	1279 (97)	1487 (97)	2766 (97)	1108 (85)
Between equator and ora serrata	38 (3)	46 (3)	84 (3)	145 (11)
Ciliary body	. . .	. . .	. . .	36 (3)
Anterior chamber angle	. . .	. . .	. . .	2 (<1)
Indeterminate	. . .	2	2	11
Not available†	. . .	30	30	862
Location of anterior tumor border				
Posterior to equator	728 (55)	757 (49)	1485 (51)	725 (57)
Between equator and ora serrata	446 (34)	604 (39)	1050 (37)	270 (21)
Ciliary body	143 (11)	172 (11)	315 (11)	167 (13)
Anterior chamber angle	. . .	. . .	. . .	55 (4)
Iris	. . .	. . .	. . .	67 (5)
Indeterminate	. . .	2	2	18
Not available†	. . .	30	30	862
Location of tumor apex relative to fovea				
Centered over fovea	9 (1)	4 (<1)	13 (<1)	32 (2)
Temporal	632 (48)	328 (40)	960 (45)	379 (29)
Superior	262 (20)	202 (25)	464 (22)	301 (23)
Inferior	255 (19)	167 (20)	422 (20)	285 (22)
Nasal	159 (12)	116 (14)	275 (13)	298 (23)
Not available†	. . .	748	748	869
Distance from closest tumor border to edge of optic disc, mm				
≤2.0	216 (16)	27 (17)	. . .	. . .
2.1-4.0	423 (32)	41 (26)	. . .	. . .
4.1-6.0	311 (24)	33 (21)	. . .	. . .
6.1-8.0	156 (12)	23 (14)	. . .	. . .
>8.0	207 (16)	35 (22)	. . .	. . .
Not available†	4	1406	. . .	. . .
Median, mm	4.0	4.5		
Closest distance between tumor and FAZ center, mm				
0	190 (15)	17 (11)	. . .	. . .
0.1-2.0	345 (26)	33 (21)	. . .	. . .
2.1-5.0	342 (26)	41 (26)	. . .	. . .
5.1-8.0	226 (17)	33 (21)	. . .	. . .
>8.0	201 (15)	35 (22)	. . .	. . .
Not available†	13	1406	. . .	. . .
Median, mm	3.0	4.5		

*Ellipses indicate not applicable or inappropriate calculation; and FAZ, foveal avascular zone.
†Requested only during early years for patients not enrolled. Missing data for enrolled patients.

patients who did not enroll (40%) and ineligible patients (29%). However, in all 3 groups of patients, more patients had tumors with the apex in the quadrant temporal to the fovea than in any of the other 3 quadrants. Among eligible patients, the tumor apex was located least often in the quadrant nasal to the fovea, owing in part to the eligibility criteria regarding tumor proximity to the optic disc.

Distance between the proximal borders of the choroidal melanoma and the optic disc and distance from the center of the foveal avascular zone to the proximal border of the tumor were reported primarily for patients who enrolled. The information was collected only through 1989 for eligible patients who did not enroll and is available for only 159 (10%) of these patients (Table 3). However, among this early group of eligible patients who did not enroll, the distance between the tumor and the optic disc borders tended to be longer than for patients who enrolled ($P=.06$). Similarly, the distance between the tumor and the center of the foveal avascular zone tended to be greater among the same group ($P=.005$). Among patients who enrolled, some part of the tumor overlay the center of the foveal avascular zone in 190 (15%).

Tumor shape or configuration was classified centrally from echograms for all patients who enrolled and for 467 patients reported during the first 3 years of patient accrual who were eligible but did not enroll (**Table 4**). The most common shape or configuration of the choroidal melanoma among eligible patients was dome, accounting for 77% of both patients who enrolled and those who did not. Collar button configurations were noted in only 207 (16%) of enrolled patients and 55 (13%) of the initial group of eligible patients not enrolled. Internal reflectivity of the tumor was added to the central assessment after patient accrual was underway; it was evaluated from baseline echograms for all patients who enrolled but for only 83 eligible patients not enrolled. Almost all of the tumors among these eligible patients were judged to have low to medium internal reflectivity; only 112 enrolled patients (9%) had internal reflectivity classified as medium-high, high, very high, or irregular. Other characteristics of the choroidal melanoma among patients who enrolled are described elsewhere.[18]

Table 4. Echographic Characteristics of Choroidal Melanoma in Eligible Patients Enrolled and Not Enrolled*

Characteristic	No. (%) of Eligible Patients: Enrolled (n = 1317)	Not Enrolled (n = 1565)
Tumor shape/configuration		
Dome	1014 (77)	327 (77)
Collar button	207 (16)	55 (13)
Lobulated or irregular	61 (5)	29 (7)
Peaked	31 (2)	9 (2)
Flat	3 (<1)	2 (<1)
Other	1 (<1)	0
Not available†	. . .	1143
Internal reflectivity		
Very low	75 (6)	8 (10)
Low	503 (39)	49 (59)
Low-medium	523 (40)	17 (20)
Medium	85 (7)	7 (8)
Medium-high, high, or very high	37 (3)	2 (<1)
Irregular	75 (6)	0
Indeterminate	19	. . .
Not available†	. . .	1482

*Ellipses indicate not applicable.
†Not assessed after November 1989 for patients who did not enroll.

VISUAL ACUITY

As given in **Table 5**, eligible patients who enrolled had worse visual acuity in the eye with choroidal melanoma but better visual acuity in the fellow eye than eligible patients who did not enroll ($P<.001$). Ineligible patients had worse visual acuity in both eyes than eligible patients ($P<.001$). More than one third (35%) of eligible patients had a visual acuity of 20/20 or better in the affected eye at the time of evaluation for this trial compared with 17% of ineligible patients. More than twice as many ineligible patients as eligible patients, 18% vs 7%, had visual acuity worse than 20/200 in the eye with choroidal melanoma.

Approximately two thirds of eligible patients but fewer than half of the ineligible patients had a visual acuity of 20/20 or better in the fellow eye (Table 5). Older age of ineligible patients only partially accounted for worse visual acuity in the fellow eye. In each age stratum, the visual acuity of fellow eyes of ineligible patients was worse than that of fellow eyes of eligible patients (data not shown).

COMMENT

Apart from slightly older ages among patients who enrolled in the COMS clinical trial of ^{125}I brachytherapy, eligible patients who enrolled and eligible patients who did not enroll had similar demographic characteristics. Choroidal melanoma of patients who enrolled were somewhat greater in apical height than those of eligible patients who did not enroll. The most frequent reason for deeming patients ineligible for enrollment was tumor proximity to the optic disc, which rendered the patient an unsuitable candidate for a radioactive plaque by COMS criteria. Thus, patients who enrolled in the clinical trial represent a subset of the target population of patients who have choroidal melanoma of the specified size.

EFFECT OF ELIGIBILITY AND EXCLUSION CRITERIA

Periodic evaluation of reasons patients were judged ineligible for enrollment led to reevaluation of exclusion criteria and some refinements and changes in eligibility criteria during the patient accrual period. Selection of an episcleral plaque with ^{125}I seeds as the radiotherapy method to be evaluated resulted in adoption of eligibility criteria specific to this delivery system, namely restrictions regarding the proximity of choroidal melanoma to the optic disc to permit the surgeon to position the plaque over the tumor base and to provide an acceptable margin for positioning errors.[29] When it became apparent after several years of patient accrual that the primary reasons for ineligibility related to proximity of the choroidal melanoma to the optic disc in otherwise eligible patients, consideration was given to several approaches to evaluating radiotherapy in comparison to enucleation for these cases. One possibility considered was to include such cases in the clinical trial in a separate stratum, with random assignment to ^{125}I brachytherapy or enucleation using a plaque designed specifically for peripapillary and juxtapapillary tumors. Concerns regarding this approach included the difficulty of plaque placement next to the optic nerve; the possibility that radiation dosimetry calculations would be incorrect owing to placement errors, and, therefore, the possibility that the tumor apex would not receive the prescribed dose of radiation; and increased risk of undetected tumor extension along the optic nerve that would not be irradiated. Thus, pooling of data across all patients assigned to ^{125}I brachytherapy would be problematic. Another possibility considered was to design a separate trial for patients who had choroidal melanoma located close to the optic disc but to use a different radiotherapy delivery system (eg, teletherapy). This approach was rejected because it was judged to be unlikely that a sufficient number of patients in this category would be available to enroll by the projected end of accrual to other COMS clinical trials. The third approach, and the one adopted, was to make no modification to the COMS protocol and to accept the fact that a sizable number of patients in the target population had been excluded from the clinical trial and that the findings from the trial could not be extrapolated reliably to such patients.

Table 5. Baseline Visual Acuity of Eligible and Ineligible Patients*

Visual Acuity	Eligible Patients: Enrolled (n = 1317)	Eligible Patients: Not Enrolled (n = 1565)	Eligible Patients: Total Eligible (n = 2882)	Ineligible Patients (n = 2164)
	No. (%) of Patients			
Eye with choroidal melanoma				
20/20 or better	413 (32)	592 (38)	1005 (35)	368 (17)
20/25 to 20/32	343 (26)	409 (26)	752 (26)	487 (23)
20/40 to 20/50	236 (18)	228 (15)	464 (16)	360 (17)
20/63 to 20/80	92 (7)	130 (8)	222 (8)	254 (12)
20/100 to 20/125	62 (5)	44 (3)	106 (4)	104 (5)
20/160 to 20/200	50 (4)	61 (4)	111 (4)	161 (8)
Worse than 20/200	110 (8)	85 (6)	195 (7)	376 (18)
Not reported	11	16	27	54
Median	20/32	20/25	20/32	20/50
Fellow eye				
20/20 or better	917 (70)	1000 (64)	1917 (67)	999 (47)
20/25 to 20/32	294 (22)	403 (26)	697 (24)	645 (30)
20/40 to 20/50	77 (6)	104 (7)	181 (6)	233 (11)
20/63 to 20/80	14 (1)	32 (2)	46 (2)	83 (4)
20/100 to 20/125	6 (<1)	7 (<1)	13 (<1)	13 (1)
20/160 to 20/200	3 (<1)	5 (<1)	8 (<1)	41 (2)
Worse than 20/200	. . .	. . .	. . .	110 (5)
Not reported	6	14	20	40
Median	≥20/20	≥20/20	≥20/20	20/25

*Ellipses indicate not applicable.

This experience has emphasized one of the difficulties encountered when designing a clinical trial for a rare condition. Before the COMS was initiated, there were no estimates from large case series of the proportion of choroidal melanoma patients in whom the proximal tumor border abutted the optic disc or extended closer than 2 mm to it. Thus, the proportion of patients who would be screened for the COMS and whose tumor location would be unsuitable for a radioactive plaque was unknown. The COMS has documented that a substantial fraction of all patients who have choroidal melanoma of a size judged suitable for brachytherapy have tumors that extend 2 mm or closer to the optic disc: 16% of eligible patients who enrolled, 17% of eligible patients who did not enroll for whom the information was reported, and 40% of patients ineligible for a single reason. Thus, more than 1000 patients in this category were evaluated for this COMS clinical trial, of whom only 216 were judged by their ophthalmologists to meet the strict eligibility criteria and enrolled.

The second most frequent reason for ineligibility of patients for the randomized trial was current or past diagnosis of another primary cancer apart from carcinoma in situ of the uterine cervix or nonmelanotic, noninvasive skin cancer. The contribution of this exclusion criterion was recognized early in the accrual period.[44] However, little consideration was given to modifying this criterion because of the potentially confounding effect on comparison of treatment arms for overall and melanoma-related mortality and because of the effects of other treatments these patients may have received.

REPRESENTATIVENESS OF COMS PATIENTS

Although patients who enrolled in this clinical trial were a selected subgroup of all patients with choroidal melanoma evaluated at COMS clinical centers, a key question is whether they were representative of patients who would be candidates for treatment with ^{125}I brachytherapy. Eligible patients who enrolled have been compared with eligible patients screened at COMS centers but who did not enroll to the extent permitted by COMS data collection policies and practices. These 2 groups of patients were found to be similar with respect to many characteristics believed to be predictive of survival outcomes but to differ with respect to tumor size. **Table 6** gives the characteristics of the choroidal tumors and of patients enrolled in this COMS clinical trial compared with those in other published studies of ^{125}I brachytherapy.[45-55] For each characteristic displayed, there was considerable variability among studies, with metrics of characteristics of COMS patients consistently contained within the distributions represented by other studies. Thus, patients who enrolled in the COMS randomized trial of ^{125}I brachytherapy vs enucleation are similar to those reported by others who underwent ^{125}I brachytherapy using an episcleral plaque. These comparisons provide additional support of the external validity of the COMS trial.

As reported, ineligible patients were older than eligible patients (Table 1). Older age among ineligible patients has been reported for other cancer trials. Hunter et al[56] reported that age criteria for eligibility and greater prevalence of comorbid conditions among older individuals resulted in decreased eligibility among cancer patients for

Table 6. Baseline Characteristics of Patients Included in Reports of Treatment of Choroidal Melanoma With Iodine 125 Brachytherapy*

		Characteristic of Patient, Tumor, or Eye							
	No. of Patients	Mean Age, y	Men, %	Right Eyes, %	Mean Apical Height, mm	Mean Longest Diameter, mm	Posterior Border Posterior to Equator, %	Anterior Border Posterior to Equator, %	Best-Corrected Visual Acuity ≥20/40, %
Enrolled in COMS	1317	59.7	50.5	50.3	4.8	11.4	97.1	55.3	68.5
Reports from comparative evaluations†									
Char et al[45]	157	58.6	...	...	5.7	10.9	...	...	...
Char et al[46]	98	58.5	...	...	5.5	10.7	...	63.3	70.4
Char et al[47]	230	...	59.1	...	5.2	...	...	58.3	70.8
Reports from case series									
Bornfeld et al[48]									
Gold shield	90	...	...	...	8.0	...	36.7	...	...
Steel shield	63	...	...	...	9.0	...	55.6	...	...
Quivey et al[49]	150	60.7	...	...	3.7	9.7	...	...	...
Fontanesi et al[50]	144	...	43.1	...	...	...	59.0	...	...
Packer et al[51]	64	...	43.8	56.2	...	...	77.0	...	...
Meyer et al[52]	40	60	40.0	50.0	...	...	60.0	...	...
Garretson et al[53]	26	58	53.8	42.3	...	...	69.2	...	69.2
Hill et al[54]	21	52.6	45.0	52.4	6.3	11.2	...	...	66.7
Kreissig et al[55]	19	58	36.8	...	5.8	11.6	100.0	26.3	...

*Ellipses indicate not reported.
†Some patients were included in more than 1 of these 3 reports.

whom a protocol was available in the Community Clinical Oncology Program. In a recent report by Hutchins et al[57] based on experience of the Southwest Oncology Group, older patients were found to be more likely to be ineligible. These investigators cited stringent eligibility criteria, coexisting medical conditions, and logistic barriers to participation of elderly patients. In the COMS, there was no upper age limit for eligibility. However, the prevalence of coexisting disease and other primary cancers was higher in older patients than younger ones. The enrolling physician considered patients' medical history and any concurrent health problems, apart from choroidal melanoma, when assessing the prognosis for 5-year survival after enrollment. Although age was to be ignored when making this assessment, some investigators reported difficulty ignoring the age of patients in their 80s and 90s, even when they were in good health. Nevertheless, only 56 patients (3%) were judged ineligible solely because they were not expected to live for at least 5 years.

VALUE OF COMPLETE CASE REPORTING

A complete accounting for all patients screened for a clinical trial has been advocated by some clinical trialists, including the authors of the CONSORT statement.[58] Such an undertaking often poses daunting logistical difficulties because it is not always simple to identify those individuals who have been screened. For example, a patient referred for screening because the target diagnosis or condition is suspected may have a different diagnosis or condition. Thus, criteria for those to be counted as screened must be established if the final accounting is to be meaningful. Few clinical trials in ophthalmology have attempted to report the total number of patients screened for the trial. Among clinical trials in cardiology, the Coronary Artery Surgery Study[59] defined the group of patients screened for the trial, described those who enrolled ("randomized"), compared them with those who were eligible but did not enroll ("randomizable"), and published survival outcomes for randomized and randomizable patients.[59,60]

Because choroidal melanoma is a rare condition, it was simple to define the patients to be reported in the COMS. As discussed earlier,[26,31,41] a decision was made during the COMS planning phase to collect demographic and clinical information for all patients referred to a participating clinical center for whom a diagnosis of choroidal melanoma was confirmed by a COMS ophthalmologist during the period of patient accrual. The primary purpose of collecting such information was to project the dates by which the target numbers of patients would be enrolled in the individual clinical trials and to estimate screening workloads at the clinical centers. However, collection of such information permitted assessment of the representativeness of the patients enrolled in COMS randomized trials and should provide substantial increases in knowledge of the epidemiologic and clinical characteristics of this primary ocular cancer.

Within the COMS multicenter organization, descriptive information has been assembled centrally for 8712 patients with choroidal melanoma who were evaluated during the 11.5-year period of patient accrual. During that time, COMS ophthalmologists examined an average of more than 700 patients with choroidal melanoma each year. Based on an estimated incidence rate of 6 to 8 cases per million population per year, these 8712 patients are believed to have comprised 35% to 40% of all of the incident cases of choroidal melanoma diagnosed in the United States and Canada during that time period. This large fraction of the choroidal melanoma patient population further supports the external validity of findings from the randomized trial. Based on known incidence rates of common eye conditions such as diabetic retinopathy and neovascular age-related macular degeneration, the proportion of all available patients who participated in the COMS is much greater than the proportion who have participated in completed clinical trials of treatments for those conditions.

In earlier reports, characteristics have been summarized for 1860 patients with large choroidal melanoma evaluated for the COMS randomized trial of preenucleation radiation vs enucleation alone[26] and for 300 patients with small choroidal melanoma evaluated for a nonrandomized prospective study.[61] In the present report, characteristics of 5046 patients with choroidal melanoma of an intermediate size judged suitable for randomization to ^{125}I brachytherapy or enucleation have been summarized. From December 1994, when accrual to the COMS clinical trial for large choroidal melanoma was completed, through July 1998, when accrual to the clinical trial of ^{125}I brachytherapy vs enucleation was completed, 635 additional patients with large choroidal melanoma were evaluated at COMS clinical centers and reported. Through July 1998, 1171 patients with small choroidal melanoma were reported to the COMS Coordinating Center. Information from all 8712 patients will be pooled and presented in future reports from the COMS Group.

GENERALIZABILITY OF FINDINGS FROM THE RANDOMIZED TRIAL

Can findings from the COMS randomized trial be extrapolated to patients who would not have met the eligibility criteria? The primary reason for ineligibility was that the tumor was located too close to the optic nerve for reliable placement of the radioactive plaque and for optimal delivery of radiation to the tumor. Mortality rates, complication rates, or both may be higher in these patients. The second most common reason for exclusion was a current or past diagnosis of another primary cancer. With progress in developing treatments that have prolonged survival and achieved apparent cure for some cancers, it may be appropriate to extrapolate findings from this trial to some patients diagnosed with choroidal melanoma with an earlier diagnosis of another cancer with no evidence of recurrence who meet all other COMS eligibility criteria. However, the COMS randomized trial will provide little or no data to support or refute the validity of this conclusion. Resources have not permitted follow-up for mortality or other outcomes among patients who did not enroll in the randomized trials. When considering treatment options, the ophthalmologist and the patient with choroidal melanoma must consider how well COMS findings are likely to apply to the individual patient based on how closely the patient meets COMS criteria for enrollment.

Through randomization to minimize bias and confounding and through attention to quality assurance and monitoring in the COMS, the randomized trials have high internal validity. External validity depends on the degree to which enrolled patients represent the population of patients with choroidal melanoma who would be candidates for both treatments evaluated. In the COMS, there were few differences in demographic or tumor characteristics between eligible patients who enrolled and those who did not that would be considered clinically and statistically significant. Furthermore, published data suggest that COMS patients who enrolled were similar to other patients with choroidal melanoma who were treated with ^{125}I brachytherapy by other physicians, further evidence to support the external validity of the trial. Therefore, findings from this COMS trial, some of which are reported in this same issue of the ARCHIVES, can be applied with confidence to patients who meet the criteria used to establish eligibility for this COMS trial.

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1998; Lisa Kailey, COA, 1996-1997; Debbie H. Lee, COA, 1990-1998; Susan N. McCart, 1991-1995; Gail A. McCoy, 1991-1995; Photographers: Mark E. Clark; Ben Baxter, 1987-1989; Bambi P. Robinson, CRA, 1991-1993; Kenneth D. Thompson, 1986-1999; Kelly K. Wilson, 1992-1996. *The Wilmer Ophthalmological Institute, Baltimore, Md:* Principal Investigator: Andrew P. Schachat, MD; Clinic Coordinators: Warren T. Doll, Jr, COA; Ellen F. Greenberg, COT; Mike Hartnett, COT; Marguerite Alexander, RN, BSN, 1986-1995; Susan Kaiser, 1988-1989; Catherine S. Sackett, RN, CANP, 1989-1993; Ophthalmologist: Nicholas T. Iliff, MD; Radiation Oncologist: Ding J. Lee, MD, PhD; Radiation Physicists: Juan F. Jackson, 1990-1998; Joseph Ianitto, 1987-1990; Wing-Chee Lam, PhD, 1987-1992; Visual Acuity Examiner: Linda T. Pollack, CO, 1988-1993; Echographer: Cathy DiBernardo, RN, RDMS; Photographers: Dennis Cain; David Emmert; Terry George, 1986-2000. *Retina Associates of Cleveland, Beachwood, Ohio:* Principal Investigators: Z. Nicholas Zakov, MD; Thomas A. Rice, MD, 1987-1997; Clinic Coordinators: Kristen Dempsey; Geraldine Daley, CRA, 1997-2000; Dina Preseren, 1988-1991; Ophthalmologists: Scott D. Pendergast, MD; Hernando Zegarra, MD; Radiation Oncologist: Donald Shina, MD; Radiation Physicists: Ned Began, MS, 1987-1998; Kunjan Pillai, MS, 1987-1998; Visual Acuity Examiners: Kimberly A. DuBois, COA; Mary A. Ilc, COT; Connie S. Keller, COA; Donna Knight, COT; Stephanie A. Schura, COA; Kathy Coreno, 1989-1994; Echographers: Pamela Rowe, CRA; Kelly Smith, CRA, COT, 1989-1994; Photographer: Sheila Smith-Brewer, CRA, COMT. *Schepens Retina Associates, Boston, Mass:* Principal Investigator: Clement Trempe, MD; Clinic Coordinator: Zena F. Nies; Ophthalmologist: John Weiter, MD; Radiation Oncologists: Mison Chun, MD, 1986-1989; David E. Wazer, MD, 1989-1994; Radiation Physicist: Kenneth Ulin, PhD, 1988-1994; Visual Acuity Examiners: Gerald Friedman, OD; Rodney Immerman, OD; Echographers: Alex Jalkh, MD; Fernando Acosta, MD, 1990-1993; Fadi Nasrallah, MD, 1987-1989; Miguel A. Quiroz, MD, 1989-1990; Photographers: Dennis C. Donovan; Thomas O'Day, 1987-1995; Andrew Rosenblum, 1986-1989; Cheryl Weber, 1986-1988. *Atlantic Eye and Face Center, Cary, NC:* Principal Investigator: Jonathan J. Dutton, MD, PhD; Clinic Coordinators: Marianne Heston, CMA, COT, 1986-1989; Cynthia Lyons, 1994-1997; Kim K. Peddle, COT, 1994-2000; Barbara A. Schuh, COT, 1990-1994; Ophthalmologist: Christopher M. DeBacker, MD, 1998-1999; Radiation Oncologists: Mitchell S. Anscher, MD; Edward Halperin, MD; Lawrence B. Marks, MD; Kenneth Leopold, MD, 1988-1993; Gustavo Montana, MD, 1987-1994; Jeffery D. Morton, MD, 1995; Gregory S. Sibley, MD, 1996-1998; Angel E. Torano, MD, 1992-1993; Radiation Physicists: Gunilla C. Bentel, RN, RTT; Thaddeus Samulski, PhD; Photographers: Beth Ann Benetz, CRA, 1989-1994; Nancy E. Knight, CRA, COA, 1991-1994; Ruth Schirmer, CRA, 1987-1997; Ophthalmic Pathologist: Alan Proia, MD, PhD, 1987-1992. *University of Virginia, Charlottesville:* Principal Investigator: Brian P. Conway, MD; Clinic Coordinators: Mary Jonni Henofer, RN, CRNO; Diana Robertson, 1989; Ophthalmologists: James S. Tiedeman, MD, PhD; Marianne Cowley, MD, 1989-1993; Eleanore M. Ebert, MD, MPH, 1993-1996; Radiation Oncologists: William C. Constable, MD; James M. Larner, MD; Michela Caruso, MD, 1992-1994; Cynthia Spaulding, MD, 1989-1992; Radiation Physicists: Alan Aqualino, PhD, 1989-1996; Gilbert D. Glennie, ME Biomed, 1995-1998; Janelle A. Molloy, PhD; Visual Acuity Examiners: James S. Chisholm, COT; Patsy Overstreet, RN, COT; Lillian L. Shoffstaff-Tyler, COT; L. Sharon Hoyle, COMT, 1992-1994; Ellen T. Murphy, AS, COA, 1993-1994; Melanie Smith, COT, 1994-1995; Karen L. Summerville, COMT, 1994-1995; Photographers: Jane G. Fleming; James Scott. *Northwestern University, Chicago, Ill:* Principal Investigators: Lee M. Jampol, MD; James Puklin, MD, 1987-1989; Clinic Coordinators: Jill M. Koecher; Beth Chiappetta, RN, 1993-2000; Gail Daubert, RN, 1987-1993; Mimi S. Mansfield, 1996-1997; Ophthalmologists: Mark Daily, MD; Gary S. Lissner, MD; Robert Schroeder, MD; David V. Weinberg, MD; Richard G. Gieser, MD, 1990-2000; Larry Wood, MD, 1991; Radiation Oncologists: Bharat B. Mittal, MBBS; William Small, Jr, MD; Ramananda Shetty, MD, 1986-1996; Radiation Physicist: Patricia Johnson, DSc, 1987-1998; Visual Acuity Examiners: Zuzanna Strugala, COA, MA; Michael Giese, OD, 1987-1991; Edgar S. Perez, OD, 1992-2000; Jeff M. Scurry, 1993-1995; Renata Swigost, 2000; Barbara A. Tallman, COT, 1991-1992; Echographers: Robert A. Levine, MD; William Myers, MD, 1986-1993; Photographers: John M. Gerty, CRA; Alexander Habib; Ned Bezinovich, 1988-1992; Karen Parque, 1988-1993; Leonard S. Richine, 1988-1998; Gloria J. Valadez, 1995-1996; Ophthalmic Pathologist: Richard O'Grady, MD, 1987-2000; Following Oncologists: Steven B. Newman, MD; John M. Shaw, MD. *University of Illinois, Chicago:* Principal Investigators: Norman P. Blair, MD; Morton F. Goldberg, MD, 1985-1989; Clinic Coordinators: Dorris Brown-Hutchins, 1988-1998; Moira Keating, 1986-1987; Christine Mullahy, COT, 1993-2000; Ophthalmologists: Steven B. Cohen, MD, 1986-1989; Kenneth I. Resnick, MD, 1990-1999; Radiation Oncologist: Edwin Liebner, MD, 1986-1995; Radiation Physicist: Richard Haas, 1987-1995; Visual Acuity Examiners: Andrew Cross, COT; Dorothe Ernest, MA, ACSW, 1989-1990; Echographers: Carolyn Anderson, MD, 1993-1995; Leonard Kut, MD, 1987-1992; Photographer: Norbert Jednock. *Cleveland Clinic Foundation, Cleveland, Ohio:* Principal Investigator: Froncie A. Gutman, MD; Clinic Coordinators: Laura Holody, COA; Susan E. Lichterman, RN; Tina E. Kiss, COT, CCRA, 1994-1997; Vivian Tanner, COT, 1987-1993; Ophthalmologists: Jill A. Foster, MD; John S. Ambler, MD, 1989-1990; John V. Linberg, MD, 1990-1991; Radiation Oncologists: Rashad El Dabh, MD, 1988-1990; Robert M. Fine, MD, 1993-1998; Jerrold P. Saxton, MD, 1990-1997; Radiation Physicists: Patrick Higgins, PhD, 1987-1994; Debbie A. Jenkins, PhD, 1993-1994; Claudio H. Sibata, PhD, 1989-1990; Visual Acuity Examiner: Susannah Hanson, CO, 1987-1997; Echographers: Karen A. King, COT; LuAnne Sculley, COMT, RDMS; Louise Berlin, 1987-1989; Cate A. Reinhard, 1993-1996; Photographers: Deborah Ross, CRA; Roberta Abram, CRA, 1986-1990; Michael P. Kelly, 1995-1999; Pamela J. Vargo, 1995-1996; Ophthalmic Pathologist: Careen Lowder, PhD, MD. *Ohio State University, Columbus:* Principal Investigator: Frederick H. Davidorf, MD; Clinic Coordinators: Cynthia S. Taylor; Jean Bildsten, RN, 1986-1990; Susan E. Cornetet, 1991-1993; Ophthalmologist: Robert B. Chambers, DO; Radiation Oncologists: Constance J. Bauer, MD; Subir Nag, MD; Sheila E. Hodgson, MD, 1987-2000; Radiation Physicists: Christos Kanellitsas, PhD; Nina Samsami, PhD; Cheng M. Su, PhD, 1989-1991; Visual Acuity Examiners: Chhanda Chaudhuri, COA; Jill D. Milliron, COA; Jerilyn G. Perry, COT, ABO; Nanci J. Cover, COA, 1992-1995; Gail Shortlidge, 1987-1995; Echographer: Paula Andrzejewska, 1992-1993; Photographers: Michael J. Keating; Scott J. Savage, EMT-A; R. Michael Clark, CRA, 1986-1990; Debra Weisenburger, CRA, 1989-1992; Following Oncologist: Gregory J. Lavalle, MD, 1997-1999. *Texas Retina Associates, Dallas:* Principal Investigator: Dwain G. Fuller, MD; Clinic Coordinators: Sally A. Arceneaux, COST, COA; Jean Arnwine; Theresa Anderson, COA, 1986-1997; Ophthalmologists: Gary Edd Fish, MD; William Snyder, MD; Radiation Oncologist: Donald Schwarz, MD; Radiation Physicists: William F. Gagnon, PhD; David B. Hammond, ABMP; Gregory McDaniel, PhD; Visual Acuity Examiners: Diane Conway, 1987-1988; Denise Dalrymple, 1987-1991; Cynthia Nork, 1987-1995; Echographers: Sandra Sue Solomon; Beth A. Pieczynski, COA, 1991-1998; Randal Straach, 1987-1995; Photographers: Hank Aguado; Bob H. Boleman; Richard E. Jones; Richard Evans, CRA, 1988-1989; John G. King, 1998-2000; Ruth Picchiottino, 1986-1987; Kevin Rose, 1988-1989; Ophthalmic Pathologist: Martha Luckenbach, MD, 1987-1989. *Porter Adventist Hospital/Centura Health, Denver, Colo:* Principal Investigator: Kenneth R. Hovland, MD; Clinic Coordinator: Suzette Compton, RN; Ophthalmologists: Douglas L. Holmes, MD; David W. Johnson, MD; Stephen T. Petty, MD; John Pope, Jr, MD; John D. Zilis, MD; Matthew W. Wilson, MD, 1997-1999; Radiation Oncologists: Seth D. Reiner, MD; David S. Shimm, MD; Wendy M. Nekritz, MD, 1997-1998; David P. Schreiber, MD, 1989-1997; Amanda J. Story, MD, 1991-1992; Radiation Physicists: Kari L. Cann, MS; Gary W. Douglas, MSc; Michael J. Bailey, MS, 1990-1996; Gregory Burns, MS, 1989-1990; J. Donald Russell, MS, 1991-1992; Echographers: David Beers; Sandra R. Sall; Photographers: Lawrence Disney, COA; Martin Lopez, COT, 1989-1990; Ophthalmic Pathologist: Robert Keyser, MD, 1989-1993. *Hermann Eye Center, Houston, Tex:* Principal Investigator: Richard S. Ruiz, MD; Clinic Coordinators: Dalia Vargas; Maria DeLaGarza, 1988-1992; Robert W.

Johnson, COA, RDO, 1992-1993; Walter Kohn, MS, 1987-1989; Georgia Triplett, OA, 1992; Ophthalmologists: Graham P. Avery, MD; Judianne Kellaway, MD; H. Michael Lambert, MD; Paul C. Salmonsen, MD; Debra J. Shetlar, MD, 1993-1994; Radiation Oncologists: Nora A. Janjan, MD; Joseph Kong, MD, 1987; William Morrison, MD, 1987-1996; Radiation Physicist: Otto Zeck, PhD; Echographers: Thomas C. Prager, PhD; Yali Zou, MD; Photographers: Charles B. Smith; Teddy J. Horton, 1987-1989. *Midwest Eye Institute, Indianapolis, Ind:* Principal Investigator: John T. Minturn, MD; Clinic Coordinators: Donna J. Agugliaro, RN, BSN; Margaret B. Kirby, RN, 1989-1992; Kimberly A. Sharkey, CST, 1989; Ophthalmologists: Ronald T. Martin, MD; William R. Nunery, MD; Radiation Oncologist: Peter Garrett, MD; Radiation Physicists: Ronald E. Berg, PhD; John Kent, MS; Photographers: Carolyn S. Lamb; William T. Bussell, 1995; Linda Huber, CRA, 1989-1995; Thomas C. Meador, 1997; Kenneth D. Phelps, 1995-1996; Ophthalmic Pathologist: Jose M. Bonnin, MD. *The University of Iowa, Iowa City:* Principal Investigator: Thomas A. Weingeist, MD, PhD; Clinic Coordinators: Connie Fountain, COT; Marcia Griffin; Ophthalmologists: H. Culver Boldt, MD; Keith D. Carter, MD; Jeffrey A. Nerad, MD; Edwin M. Stone, MD, PhD; Christopher F. Blodi, MD, 1987-1992; Radiation Oncologists: David H. Hussey, MD; Antonio Vigliotti, MD, 1987-1991; B-Chen Wen, MD, 1991-2000; Radiation Physicists: Womah S. Neeranjun, MS, 1994-1997; Edward Pennington, ScM, 1987-2000; Echographers: Tom Fisher; Laura Warner; Karl C. Ossoinig, MD, 1986-1997; Photographers: Edward Heffron, MA; Stefani V. Karakas, CRA; Randall E. Verdick; Ophthalmic Pathologist: Robert Folberg, MD, 1986-2000; Following Oncologists: Gerald H. Clamon, MD; Susan A. Kambhu, MD, 1996-1998. *Estelle Doheny Eye Foundation, Los Angeles, Calif:* Principal Investigators: A. Linn Murphree, MD; Peter E. Liggett, MD, 1986-1991; Stephen J. Ryan, MD, 1985-1986; Clinic Coordinators: Margaret Padilla; A. Frances Walonker, CO, COMT; Beth Quillen-Thomas, COMT, 1987-1991; Ophthalmologists: Kenneth R. Diddie, MD; Jennifer I. Lim, MD; Leroy C. McNutt, MD; Brent C. Norman, MD; Donald A. Frambach, MD, 1991-1998; Sanjay Logani, MD, 1998-1999; Sunil S. Patel, MD, PhD, 1997-1998; Michael A. Singer, MD, 1993-1997; Radiation Oncologists: Deirdre Cohen, MD; Zbigniew Petrovich, MD; Daphne Palmer, MD, 1988-1990; Radiation Physicists: Melvin A. Astrahan, PhD; Gary Luxton, PhD; Visual Acuity Examiner: Maria Trujillo, 1987; Echographer: Ronald L. Green, MD; Photographers: Tracy Nichols, CRA; Stephen Delgado, 1987-1990; Jan Johnson, 1987; Michon Rozier, 1987-1989; Mark G. Williams, CRA, 1990-1993. *Jules Stein Eye Institute, Los Angeles, and Southern California Permanente Group, Panorama City:* Principal Investigators: Robert E. Engstrom, Jr, MD; Bradley R. Straatsma, MD, 1985-1995; Clinic Coordinators: Robert Almanzor; Robert Caldwell, PA; Karen Baranick, COA, 1993-1995; Margaret G. Haslop, COA, 1987-1989; Barbie S. Pinaire, COA, 1990; Ophthalmologists: Man M. Singh Hayreh, MD; Hector Sulit, MD; Radiation Oncologists: Robert G. Parker, MD; C. Michelle Burnison, MD, 1992-1999; Michael Selch, MD, 1987-1989; Radiation Physicists: James Smathers, PhD; Robert E. Wallace, PhD; Visual Acuity Examiners: Lisa A. Barnhart, OD; Melissa Chun, OD; Burton Krell, OD, 1988-1995 (deceased); Echographers: Marci L. Fishman, RDMS; Barry Kerman, MD; Joel N. Moral, COT; Nancy London, COA, 1986-1996; Ron Smith, COT, 1991-1993; Lynn Wynbrandt, 1987-1989; Photographers: Michael Heneghan; Dennis Thayer; Don Allen, 1989-1990; Anne P. Bolton, 1991-1995; Orly Catz, 1996-1997; Joan Greenspan, 1988-1989; Ophthalmic Pathologists: Ben J. Glasgow, MD; Robert Y. Foos, MD, 1987-1994. *University of Wisconsin, Madison:* Principal Investigator: Suresh R. Chandra, MD; Clinic Coordinators: Margo S. Blatz; Jennie R. Perry; Betty Lewis, 1987-1998; Lyn Stewart, 1987-1990; Ophthalmologists: Justin L. Gottlieb, MD; Bradley Lemke, MD; Mark J. Lucarelli, MD; Richard Dortzbach, MD, 1986-1998; Frank Myers, MD, 1987-1998; Timothy W. Olsen, MD, 1997-1998; Thomas Stevens, MD, 1986-1995; Radiation Oncologists: Minesh P. Mehta, MB ChB/ACR; Richard Steeves, MD, PhD; Radiation Physicists: Bhudatt R. Paliwal, PhD; Bruce Thomadsen, PhD; Siamak Shahabi, PhD, 1987-1991; Visual Acuity Examiners: Jim N. Henry, COT; Guy Somers, RN; Helen Lyngaas Myers, 1993-1998; Diane Quackenboss, RN, 1987-1995; Echographer: Laura Kassel, COA; Photographers: Robert Harrison; Gene Knutson; Hugh D. Wabers; Michael Neider, 1987-1995. *Bascom Palmer Eye Institute, Miami, Fla:* Principal Investigators: Timothy G. Murray, MD; Don H. Nicholson, MD, 1985-1987; Clinic Coordinators: Nicole Cicciarelli, CST; Joanka M. LoBracco; Madeline L. Del Calvo, 1992-1995; Marianela Castellanos, 1987-1994; Ophthalmologists: Mary Lou Lewis, MD, 1990-1995; Patrick E. Rubsamen, MD, 1991-1995; Radiation Oncologists: Arnold M. Markoe, MD, ScD; Beatriz E. Amendola, MD, 1990-1991; Bruce Horowitz, MD, 1987-1988; Abdon Medina, MD, 1988; James Schwade, MD, 1988-1994; Radiation Physicists: Jeffrey A. Fiedler, MSc, 1992-1996; Pavel Houdek, PhD, 1987-1995 (deceased); Vincent J. Pisciotta, MS, 1988-1995; Xiaodong Wu, PhD; Visual Acuity Examiners: Nury Delgado; Rita D. Swoopes; Edward Attaway, OD, 1987-1988; Steven C. Deen, OD, 1988-1992; Donald R. Faimon, OD, 1990-1992; Mailyn Gonzalez, 1993-1997; Steven Holbrook, OD, 1987-1988; David J. Kowal, OD, 1988-1990; Mark Liebetreu, OD, 1988-1989; Mark D. McClintock, OD, 1988-1990; Esther Pastrana, COT, 1994-1996; Nidia Y. Rosado-Guzauskas, 1996-1999; Deyanira I. Serrano, 1992-1993; June Thompson, 1993-1994; Echographers: Fiona Ehlies; J. Randall Hughes; E. Kym Gendron, 1987-1999; Photographers: Ditte Hess; Isabel Rams, CRA; Ross Jarrett, 1988-1992; Ophthalmic Pathologists: Victor T. Curtin, MD; Robert H. Rosa, Jr, MD; Following Oncologist: Lynn G. Feun, MD. *Medical College of Wisconsin, Milwaukee:* Principal Investigators: Judy E. Kim, MD; Gary W. Abrams, MD, 1985-1986; William F. Mieler, MD, 1986-1999; Clinic Coordinators: Sharon A. Rekow; Mary Andrus, 1999; Prudence Helt, LPN, 1986-1988; Ophthalmologist: Gerald J. Harris, MD; Radiation Oncologists: Beth Erickson, MD; Maurice Greenberg, MD, 1986-1991; Radiation Physicists: Michael T. Gillin, PhD; Darwin L. Zellmer, PhD, 1991-1995; Visual Acuity Examiner: Dale T. Jurkiewicz, 1989-1993; Echographer: Michael Lewandowski, RDMS, 1986-1996; Photographers: James R. Phillips, AS; Walter Wipplinger; Mary Nauertz-Ritchie, 1987-1991; Jim L. Walters, 1989; Ophthalmic Pathologist: James Caya, MD, 1986-1988. *Montreal Hospitals, Montreal, Quebec:* Principal Investigator: Christine Corriveau, MD; Clinic Coordinators: Liette Bousquet, COA; Suzanne Casavant, RN; Margaret Gagnon, RN, 1995-1998; Darlene O'Connor, 1989-1993; Ophthalmologists: Mikael A. Sebag, MD; Jean Deschenes, MD, 1989-1998; Radiation Oncologists: Michel Gelinas, MD; Pierre Rousseau, MD; Claudine Theriault, MD, 1989-1993 (deceased); Mariam Yassa, MB, BCH, 1989-1998; Radiation Physicists: Noel Blais, PhD; Michael D. C. Evans, MSc, 1990-1998; Visual Acuity Examiners: Claire Choquette, OC; Deborah M. Williams, COT, 1990-1998; Echographers: Magdi M. Mansour, MD, MSc, 1993-1998; Diane Chialant, RN, 1989-1992; Photographers: Marc J. Blouin, OA; Rosario Bate, BN, CRA, COT, 1990-1992; Lilianne L. M. Marcon, COA, 1993-1998; Ophthalmic Pathologists: Guy S. Allaire, MD; Seymour Brownstein, MD, 1990-1992. *Touro Infirmary, New Orleans, La:* Principal Investigator: Gerald Cohen, MD; Clinic Coordinators: Sandra W. Palao, CST; Susy A. Steib, 1989-1992; Radiation Oncologist: Mary Ella Sanders, MD; Radiation Physicists: Lawrence J. Metevier, MS; Donald G. Jackson, 1990-1992. *Tulane University, New Orleans, 1989-1995:* Principal Investigator: Barrett Haik, MD; Clinic Coordinators: Elaine M. Eckols, COA: Carolyn A. Bushaw, RN, 1989-1991; Natalie Loyacano, 1993; Ophthalmologist: Zeynel A. Karcioglu, MD, 1989-1993; Visual Acuity Examiner: Alan Faulkner, MD; Echographer: Mary E. Smith, RDMS, 1990-1995; Photographer: Rebecca Thiele, CRA. *Cornell University, New York, NY:* Principal Investigator: David H. Abramson, MD; Clinic Coordinators: Diane M. Murphy; Indira S. Rollins, CST; Katherine S. Budinger, COA, 1995-1997; Stewart P. Martin, CRA, 1993-1994; Agnese Romanella-Weil, RN, BSN, CRNO, 1989-1992; Camille A. Servodidio, RN, MPH, CRNO, 1986-1997; Ophthalmologist: Mary E. Mendelsohn, MD, 1994-1996; Radiation Oncologists: Beryl McCormick, MD; Daniel Fass, MD, 1987-1993; Karen D. Schupak, MD, 1993-1997; Radiation Physicists: Patrick J. Harrington, 1990-1999; Soutung Chiu-Tsao, PhD, 1987-1990; Photographers: Dorothy A. Fong, COT; Rob Wolfson, CRA; Ophthalmic Pathologists: Winston G. Harrison, MD; Sarah S. Frankel, MD, 1990-1991. *New York Eye and Ear Infirmary, New York, and North Shore University Hospital, Manhasset, NY:* Principal Investigator: Paul T. Finger, MD; Clinic Coordinators: Jeannie Ryan, COT; Paula A.

Muir, 1997-1999; Cynthia Pearlberg, 1987-1996; Ophthalmologist: Samuel Packer, MD; Radiation Oncologists: Anthony M. Berson, MD; Jay Bosworth, MD; Daniel DeBlasio, MD, 1988-1991; Louis Potters, MD, 1992-1994; Radiation Physicists: Alfonso Buffa, MS; Tony Ho, MS, 1987-1991; Willie Vahid Jalayer, PhD, 1991; Nokul Panigrahi, PhD, 1994-1999; Richard Riley, PhD; Andrzej Szechter, PhD; Visual Acuity Examiners: Joseph Brendel, COT; Lori Reminick, OD, 1997; Photographers: Nancy Gonzalez, COA; James W. Perry; Michael McCray, 1987-1989; Roy E. Tuller, RBP, 1989-1990; Jon Wilder, 1990-1999; Ophthalmic Pathologist: Steven A. McCormick, MD. *McGee Eye Institute, Oklahoma City, Okla:* Principal Investigator: Reagan H. Bradford, Jr, MD; Clinic Coordinators: Lisa Ogilbee, COA; Jamie Johnson, LPN, 1999; Janie M. Shofner, COA, CCRA, 1990-1998; Ophthalmologists: P. Lloyd Hildebrand, MD; Sumit K. Nanda, MD; Scott C. Sigler, MD; Radiation Oncologists: Robert C. Gaston, DO, 1997-1999; Elizabeth J. Syzek, MD, 1995-1998; Radiation Physicists: B. Wally Ahluwalia, PhD; Echographer: Jaime Rodriguez; Photographer: Russell D. Burris, CRA, COT. *Wills Eye Hospital, Philadelphia, Pa, 1992-1995:* Principal Investigator: Jerry A. Shields, MD; Clinic Coordinators: Diane McDonald, COT; Margaret C. Morley, 1992-1993; Ophthalmologists: Patrick V. De Potter, MD; Carol L. Shields, MD; Radiation Oncologist: Luther W. Brady, MD; Radiation Physicists: Lester Tripp, MSc; Patrick Glennon, 1992; Echographer: Queen Warrick, COA; Photographer: Matthew Weiner, CRA; Ophthalmic Pathologist: Ralph C. Eagle, Jr, MD. *Retina-Vitreous Consultants, Pittsburgh, Pa:* Principal Investigators: Karl R. Olsen, MD; Jeffrey S. Rinkoff, MD, 1988-1996; Clinic Coordinators: Donna L. Green, RN, CRNO; Barbara J. Bahr, LPN, 1988-1999; Ophthalmologists: Randall L. Beatty, MD; Robert L. Bergren, MD; Gary Weinstein, MD; Scott R. Hobson, MD, 1991-1994; Radiation Oncologist: Melvin Deutsch, MD; Radiation Physicists: Kurt G. Blodgett, MS; Andrew Wu, PhD, 1988-1994; Virgil E. Yoder, MS, 1988-1989; Visual Acuity Examiners: Lynn A. Wellman; Linda A. Wilcox, COA; Kimberly S. Flook, COA, 1992-1993; Catherine Jaborsky, 1988-1989; Beverly J. Keenen, 1991-1992; Photographers: Alan F. Campbell, CRA; David Steinberg, CRA; Gary Vagstad, CRA; Mary M. Good, 1993; Kim A. Mellinger, COA, 1991. *Casey Eye Institute and Devers Eye Clinic, Portland, Ore:* Principal Investigators: David J. Wilson, MD; Robert C. Watzke, MD (Casey), 1986-1990; Richard G. Chenoweth, MD (Devers), 1987-1996; Clinic Coordinators: Shirley Ira, COT; Anna Jane Leigh, COT, 1989-1992; Elaine Lorimer, 1987-1989; Ophthalmologists: Roger A. Dailey, MD; Richard F. Dreyer, MD; Diana L. Habrich, MD; Michael Klein, MD; Colin Ma, MD; John L. Wobig, MD; Frederick Fraunfelder, MD, 1987-1995; Radiation Oncologists: Harper D. Pearse, MD; Kenneth R. Stevens, MD; Juan V. Fayos, MD, 1989; Radiation Physicist: Raymond Fry, MS; Visual Acuity Examiners: Julie Arends, COMT, 1987-1995; Lori Rusch, 1987-1989; Echographers: Mark Evans; Patrick B. Rice, CRA, 1991-1997; Thomas Talbot, MD, 1986-1991; Photographers: Howard G. Daniel, 1990-1995; Fernando Corrada, 1988-1989. *Mayo Foundation, Rochester, Minn:* Principal Investigator: Dennis M. Robertson, MD; Clinic Coordinators: Margaret J. Ruszczyk, CCRA; Mary Schumann, COMT; Ophthalmologist: Helmut Buettner, MD; Radiation Oncologists: Ivy A. Petersen, MD; Paula Schomberg, MD; Scott L. Stafford, MD; John D. Earle, MD, 1986-1996; Radiation Physicists: Robert W. Kline, PhD; Peter Yeakel, RTT; Visual Acuity Examiners: Barbara Eickhoff, COMT; Deborah Miller, COMT; Photographers: Thomas Link; Jay Rostvold; Ophthalmic Pathologist: R. Jean Campbell, MD, 1988-1998; Following Oncologists: Kurt W. Carlson, MD; Taylor J. Hays, MD. *Associated Retinal Consultants, Royal Oak, Mich:* Principal Investigator: Raymond R. Margherio, MD; Clinic Coordinators: Kristi L. Cumming, RN, MSN; Patricia Manatrey, RN; Sheri McCurley, RN, BSN, 1986-1989; Beth Mitchell, RN, 1989-1998; Ophthalmologists: Antonio Capone, Jr, MD; Bruce R. Garretson, MD; Alan R. Margherio, MD; Michael T. Trese, MD; Patrick L. Murphy, MD, 1987-1995; Radiation Oncologists: Alvaro Martinez, MD; Richard Matter, MD; Radiation Physicists: Gregory Edmundson, RT; Elizabeth C. Mele, MS; Nancy R. Rizzo, MS, 1992; Visual Acuity Examiner: Virginia S. Regan, RN, BSN; Echographers: Laurie K. Lau-Sickon, RDMS; Andrea P. Nielsen, OD; William Park, OD, 1987-1998; Photographers: Craig Bridges, CRA; Frances McIver; Patricia Streasick, CRA; Lynette D. Szydlowski, COT; Rachel E. Falk, 1997-2000; John L. Johnson, 1992-1994; Jeffrey Sobel, CRA, 1987-1990; Deborah Sorenson, 1987-1992; Following Oncologists: David A. Decker, MD; Ishmael Jaiyesimi, MD. *Washington University, St Louis, Mo:* Principal Investigators: J. William Harbour, MD; Lucian Del Priore, MD, 1995-1998; Morton E. Smith, MD, 1986-1995; Clinic Coordinators: Mary V. Gaines; Lori Ann Clark, COT, 1994-1996; Floyce Scherrer, RN, 1987; Ophthalmologists: Philip L. Custer, MD; Henry J. Kaplan, MD, 1989-2000; Lawrence Schoch, MD, 1987-1989; Radiation Oncologists: Venkata Devineni, MD, 1987-1994; Hsiu-San Lin, MD, PhD, 1989-1995; Radiation Physicist: Eric Slessinger, MS, 1987-1994; Echographer: William Hart, MD; Photographer: Rhonda Curtis, COT. *University of Texas, San Antonio:* Principal Investigator: Wichard van Heuven, MD; Clinic Coordinators: Leticia S. Leija, RDMS; Judy Attaway, 1987-1991; Debbie A. Schifanella, 1991-1993; Carolyn G. Torres, LVN, 1993-1996; Ophthalmologists: Steven Chalfin, MD; Gary Cowan, MD, 1986-1988; David E. Holck, MD; Lt Col Donald A. Hollsten, MD; Lina M. Marouf, MD; Kenneth L. Piest, MD; James W. Speights, MD; Randall Beatty, MD, 1990-1991; Bailey L. Lee, MD, 1995-1999; John Shore, MD, 1986-1988; Christopher T. Westfall, MD, 1992-1995; Radiation Oncologists: Ardow Ameduri, MD; Carlos J. Hernandez, MD; Anthony J. Campbell, MD, 1993-1995; John J. Feldmeier, DO, 1988-1992; Radiation Physicists: James M. Hevezi, PhD; Robert G. Waggener, PhD, 1987-1995; Visual Acuity Examiners: Leticia G. Curup, COT; Syed Z. Parvez, MD; Karlyn Gansel, OD, 1989-1990; Steve L. Hand, OD, 1990-1992; Echographers: Barry M. Byrne, 1987-1989; Jacalyn McAdam, ARDMS, 1990-1993; Photographers: Paul Comeau, CRA; Richard Sabo, CRA, 1988-1995; Ophthalmic Pathologists: William C. Lloyd III, MD; Andrew Lawton, MD, 1986-1995. *Schatz, McDonald, Johnson, and Ai, San Francisco, Calif:* Principal Investigators: Robert N. Johnson, MD; Howard Schatz, MD, 1989-1991; Clinic Coordinators: Nubia E. Garcia, RN; Karen Laughlin, LVN; Renee Y. Guertin, 1994-1997; Anne Miller, 1990-1994; Radiation Oncologists: Sara M. Huang, MD; Jerold P. Green, MD, 1988-1991; John Meyer, MD, 1987-1995; Abhinand Peddada, MD; Radiation Physicists: Avelino C. Dimaya, 1991-1999; Devorah Novack, MS; Visual Acuity Examiners: Kevan Curren; Marsha A. Apushkin, MD, 1999-2000; Marian M. D'Angelo, RN, 1995-1997; Jill Teachout, COA, 1989-1990; Echographer: Margaret M. Stolarczuk, OD; Photographers: John Uy; Michael Kelly, 1988; Karen Kline, 1988-1989; Michael McKenzie, 1989-1994; Irina A. Rozenfeld, 1995-2000; Scott M. Wild, 1994-1996. *Ophthalmic Consultants Northwest, Seattle, Wash:* Principal Investigator: Edward B. McLean, MD; Clinic Coordinators: Shiela M. Davidson; Tuija H. Kaarrekoski, 1995; Pauline M. Manner, 1996-1998; Marilyn M. Teeter, MA, 1989-1994; Ophthalmologist: J. Timothy Heffernan, MD; Radiation Oncologist: James R. Eltringham, MD, 1989-1997; Radiation Physicists: Juri Eenmaa, PhD; Douglas Schumacher, 1989-1994; Visual Acuity Examiner: Ralph Archer, COMT, 1989-1995; Photographers: Robin Wilkins, CRA; Phillip R. Goedert, 1995; Ophthalmic Pathologist: Bruce G. Kulander, MD. *University of Washington, Seattle:* Principal Investigator: Craig G. Wells, MD; Clinic Coordinator: Betty Lawrence, COA; Ophthalmologist: Robert E. Kalina, MD; Radiation Oncologists: Wui-Jin Koh, MD; Karen Lindsley, MD; Radiation Physicists: Stan Brossard, MS, 1990-1992; Mark W. Davidson, MS, 1992-1993; Alina Popescu, PhD; Karen M. Singer, MS; Photographers: Brad Clifton; Ron Jones. *Retina Associates of Florida, Tampa:* Principal Investigator: W. Sanderson Grizzard, MD; Clinic Coordinators: Janet Traynom, COT; Beverly B. Colding, LPN, 1994; Mark A. Greenwald, 1992; William J. Malenfant, 1996-1999; Sandra K. Myers, COT, 1989-1992; Theresa L. Shannon, COA, 1994-1995; Ophthalmologists: Mark E. Hammer, MD; Peter R. Pavan, MD; Steven M. Cohen, MD, 1995-1997; Radiation Oncologist: William Assad, MD; Radiation Physicists: Ignacio A. Ferras III, MS; Ashoka Bhargava, MS, 1993-1996; Zoubir Ouhib, MS, 1989-1992; Visual Acuity Examiner: Jayne Craig; Photographer: Janice M. Lightbourn, 1993-1997; Ophthalmic Pathologist: Curtis E. Margo, MD; Following Oncologist: Curtis C. Adolphson, MD. *Scott and White Memorial Hospital, Temple, Tex:* Principal Investigator: J. Paul Dieckert, MD; Clinic Coordinators: Janice M. Jeske, RN; Juanice R. Hamm, RN, 1987-1995; Suzann M. Whatley, COT, 1988-2000; Ophthalmologist: Glen O. Brindley, MD; Radiation Oncologist: Odis D. Skinner, MD, 1987-1991; Radiation Physicist: Louis H. Deiterman, PhD, 1988-1998; Visual Acuity Examiners: Terry L. Walters, COT, 1989-2000; Donald B. Skelton, COMT, 1987-1994; Echographer: Vernon M. Hermsen, MD, 1988-1989; Photographers: George L. Methvin; Barry McMickle, 1987-1989. *Ontario Cancer Institute, Toronto:* Principal Investigators: E. Rand Simpson, MD; Peter Fitzpatrick, MBBS, 1986-1989; Clinic Coordinators: Lee Penney, COA; Andrea Mackie, COA, 1987-1994; Ophthalmologists: Qazi Ahmad, MD; Hugh McGowan, MD; James H. Oestreicher, MD; Herbert Tanzer, MD; Murray Christianson, MD, 1987; Manijeh Contractor, MD, 1994-1998; David A. Ewing-Chow, MD, 1992-1994; Martin Kazdan, MD, 1987-1992; Shyam Mohan R. Kodati, MD, 1991-1992; Hesham Lakosha, MD, 1998-1999; John A. McWhae, MD, 1990-1991; Andrew Mis, MD, 1995-1997; Katherine Paton, MD, 1988-1989; Radiation Oncologists: Normand J. Laperriere, MD; David Payne, MD; Radiation Physicists: Barbara Japp; Sandra Ahonen, RTT, 1989-1993; Ivan Yeung, PhD; Echographer: Charles J. Pavlin, MD; Photographers: Allan Connor; Karen M. Hahn, CRA; Janet Hammond, RN, 1989-1991. *Retina Associates Southwest, Tucson, Ariz:* Principal Investigator: Leonard Joffe, MD; Clinic Coordinators: Mari L. Bunnell, COA; Sally K. Brandon, LPN, COT; Pamela Alcala, COA, 1989-1990; Christine Hanson, RN, BSN, 1990-1994; Lois J. Loescher, MS, RN, 1989-1990; Ophthalmologists: Denis M. Carroll, MD; Reid F. Schindler, MD; Radiation Oncologists: J. Robert Cassady, MD, 1989-1995; Helen Fosmire, MD, 1989-1997; James R. Oleson, MD, PhD, 1997-1999; Terence J. Roberts, MD, 1992-1993; David Shimm, MD, 1990-1997; Patrick Swift, MD, 1990; Radiation Physicists: Chee-Wai Cheng, PhD, 1991-1999; Bruce Lulu, PhD, 1989-1997; Wendell Lutz, PhD, 1990-1993; Jeffrey Williamson, 1989; Visual Acuity Examiner: Elizabeth S. Weimann, COT; Echographer: Laura S. Rodgers, COA; Photographers: Larry M. Wilson; Darryl C. Freilinger, CRA, 1990-1991; Stacey A. Halper, 1995-1997.

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Scheie Eye Institute, Philadelphia, Pa, Chairman's Office: COMS Group Chair: Stuart L. Fine, MD; Assistants: Donna DiSciullo; Gretchen E. Goller; Emily Fatula, 1993-1995; Lee McCaffrey, MA, 1987-1998; Virginia Nido, 1991-1993; Alissa M. Reich, MSEd, 1995-1997; Jane Ebert Torano, 1985-1988; Wendy Tucker, 1998; Administrative and Support Personnel: Cheryl Hurley, 1991-1997; Carol Lackey, 1997; Debbie Russo, 1991-1992. *The Wilmer Ophthalmological Institute, Baltimore, Md, Coordinating Center:* Principal Investigator: Barbara S. Hawkins, PhD; Study Statistician: Marie Diener-West, PhD; Study Epidemiologist and Protocol Monitor: Claudia S. Moy, PhD; Jan A. Markowitz, PhD, 1985-1991; Diane M. Willis, PhD, 1989-1991; Central Data Coordinators: Reneé Y. Mobley; Townes C. Coates, 1989-1997; Wanda S. Jeffreys, 1988-1989; Joyce L. Keenan, 1990-2000; Donna C. Optican, MAS, 1985-1987; Clinic Monitors: Rosemary A. Hollick, MS; Kelly S. Manos, MAS, 1991-1998; Christopher Kane, MS, MPH, 1994-1995; Biostatisticians: Marta J. Marsh, MS; B. Michele Melia, ScM; Statistical Programmers: Sandra M. Reynolds, MA; Craig E. Belongie, 1991-1994; Brett A. Hoover, 1990-1991; Kimberly L. King, 1994-1996; Päivi H. Miskala, MSPH, 1996-1998; Systems Analysts and Programmers: M. Marvin Newhouse; Paul B. Connor, 1988-1992; Stephen C. Grubb, MS, 1985-1996; Jonathan S. Kerman, 1990-1999; Nimish R. Shah, 1994-1995; Administrative and Support Personnel: Patricia A. James; Lisa A. Lassiter; Nancy A. Prusakowski, MS; Gregory Surplus; George T. Allen, 1989-1993; Amanda Davis, 1987-1990; Irene L. Felicetti, 1985-1998; Darryl M. Harper, 1989-1991; Takisha R. Kiah, 1995-1998; Darren T. Otero, MM, 1995-1998; Jeffrey D. Silverman, 1998-1999; Katasha Smart-Davis, 1992-1994; Dawn K. Smith, 1997-1999; Juanchulla Terry, 1990-1992. *Mars Hill College, Mars Hill, NC, Echography Center:* Principal Investigator: Sandra Frazier Byrne; Coordinators: Linda Schmidt; Valerie Bruns, 1999-2000; Michelle Comella, 1986-1987; Jeanette Moreno, 1992-1994; Linda Pritchard, 1996-1999; Pat Superfine-Rivera, 1987-1992; Florence Zerbarini, 1994-1996; Graders: E. Kym Gendron, 1985-1994; J. Randall Hughes, 1985-1994; Maria Rivera, 1993-1994; Cathy DiBernardo, RN, 1985-1991. *University of Wisconsin, Madison, Pathology Center:* Principal Investigator: Daniel M. Albert, MD; Coordinators: Michelle Zimbric; Tracy Perkins, MPH, COA, 1998; Nancy L. Robinson, 1986-1998; Ophthalmic Pathologists: Thaddeus Dryja, MD, 1986-1992; Guillermo de Venecia, MD, 1986-1999; Ingolf Wallow, MD, 1986-1998; Technical and Support Personnel: Lorie Buckley, 1992-1994; Emily Hajar, 1989-1992; Bartholomew Johnson, 1989-1992. *The University of Iowa Hospitals and Clinics, Iowa City, Photograph Reading Center:* Principal Investigators: H. Culver Boldt, MD; Thomas A. Weingeist, MD, PhD, 1986-1999; Computer Consultant: Paul R. Montague; Coordinators: Luke P. Groben; Diane Anderson, 1992-2000; Laurie Smith, 1988-1992; Judy Strauss, 1986-1988; Ophthalmologists: James C. Folk, MD, 1988-1992; Edwin M. Stone, MD, PhD, 1993-1999; Photograph Readers: Rosemary Brothers; Ivy Andersen, 1997-1998; Cheryl J. Hiner, 1998-1999; Ray Northway, 1986-1991; Luan Streb, 1989-1992; Christine Taylor, 1991-1993; Carolyn Vogel, 1986-2000; William Ward, 1994-1996. *M. D. Anderson Hospital, Houston, Tex, Radiological Physics Center:* Principal Investigator: William F. Hanson, PhD; Physicists: David Followill, PhD; Connel Chu, MS, 1995-1996; Thomas Kirby, PhD, 1986-1988; Carter B. Schroy, PhD, 1989-1991; Ramesh C. Tailor, PhD, 1992-1994; Technical Support Personnel: Elizabeth Engelberg; Cathy Kasper, 1987-1991; Paula Kennedy, 1986-1996.

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National Eye Institute, National Institutes of Health, Bethesda, Md: Program Staff: Natalie Kurinij, PhD; Jack A. McLaughlin, PhD, 1986-1989; Richard L. Mowery, PhD, 1990-1994.

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